

Person 1 — Project Lead, System Architecture & Integration

Intelligent Land Record Digitization, Validation & Urban Cadastral Mapping Platform |
Step-by-step implementation specification | Unified Master Blueprint v2.0

Assigned role: Person 1 — Project Lead, System Architecture & Integration

Document set: Unified Master Blueprint v2.0

Scope: SIH 12 + SIH 18 combined platform

Canonical rule: This document must not introduce a technology, endpoint, role, state, table, or handoff that conflicts with the Master Blueprint.

1. Mission

Own the control plane, shared contracts, workflow orchestration, integration events, cross-module compatibility, integration tests and final release process. Do not implement specialist ML algorithms; make every specialist module callable, observable and testable.

2. Non-negotiable shared rules

- Use the canonical stack, API paths, entities, statuses, application roles and event names from Master Blueprint v2.0.
- Do not change another person's contract by silently renaming fields or endpoints.
- All work is project-scoped through `project_id`.
- All long-running work is asynchronous with job/workflow status.
- Original documents/source imagery remain immutable.
- Every AI output carries confidence, provenance, `model_version` and timestamp.
- Every critical mutation is auditable.
- Before handoff, run tests and provide evidence.

3. Exact handoff principle

A handoff is complete only when the receiving person can consume the documented output without guessing field names, endpoint names, status values, storage locations, or semantics.

4. Repository ownership

```
backend/app/api/v1/  
backend/app/services/workflow/  
backend/app/events/  
backend/app/schemas/  
docs/api/  
docs/architecture/  
tests/integration/  
release evidence / checklists
```

5. Step-by-step implementation plan

Step Create the repository skeleton from the Master Blueprint. Verify every P1-owned path exists.

Step Create the canonical API contract document and freeze method/path/purpose for all endpoint groups.

Step Create the canonical workflow state enum: UPLOADED, QUEUED, PROCESSING, EXTRACTED, VALIDATING, REVIEW_REQUIRED, VALIDATED, PUBLISHED, FAILED.

Step Implement a state-transition function that rejects illegal transitions and records the actor/job context.

Step Implement `workflow_runs` and `workflow_steps` with progress, stage history, errors, start/finish timestamps.

Step Implement document workflow orchestration in the exact sequence from the Master Blueprint.

Step Implement cadastral workflow orchestration in the exact sequence from the Master Blueprint.

Step Implement integration event persistence and delivery tracking.

Step Define event schemas for `document.ocr.completed`, `validation.completed` and `geoai.completed`.

Step Implement idempotency at workflow start and retry boundaries so repeated requests cannot create duplicate runs.

Step Implement retry-from-failed-stage behavior while preserving previous stage history.

Step Implement cancellation only for states where cancellation is valid; never fake cancellation of completed work.

Step Expose workflow progress and stage history through the canonical endpoints.

Step Create integration contract tests that simulate P2/P3/P4 outputs and verify the orchestrator advances correctly.

Step Create cross-project authorization integration tests with P5.

Step Create final release checklist with evidence for tests, migrations, security and demo.

6. Canonical orchestration behavior

Document:
upload -> preprocess -> OCR -> extraction -> validation -> review -> approval -> publication

Cadastral:
imagery upload -> CRS validation -> preprocessing -> segmentation
-> polygonization -> topology validation -> GIS review -> publication

Failure:
failed stage -> FAILED -> QUEUED -> retry from failed stage

Every stage:
1. validate input contract
2. create/update workflow_steps
3. invoke service
4. persist output reference
5. emit integration event
6. update workflow status
7. create audit event

7. Integration event contracts

Event	Required payload	Consumer
document.ocr.completed	document_id, page_ids, model_version_id, result_uri	P1/orchestrator + P2/P3
validation.completed	document_id, result_count, critical_error_count, review_task_ids	P1/P6
geoai.completed	project_id, layer_ids, feature_counts, topology_error_count	P1/P6

8. Required API behavior

Endpoint	Requirement
/api/v1/projects/{project_id}/workflows	Start document or cadastral workflow after authorization
/api/v1/workflows/{workflow_run_id}	Return status, current_stage, progress and stage history
/api/v1/workflows/{workflow_run_id}/retry	Retry only failed/retryable work
/api/v1/workflows/{workflow_run_id}/cancel	Cancel queued/pending work
/api/v1/integrations/events	Expose failed event deliveries for recovery

9. Integration tests

- Create a project and assign application roles without confusing them with P1-P6 engineering ownership.
- Start a land-record workflow and verify every transition.
- Start a cadastral workflow and verify every transition.
- Force a worker failure; retry and confirm no duplicate result.
- Repeat an idempotent start request; confirm the same workflow is returned.
- Cancel queued work and confirm terminal behavior.
- Verify unauthorized project access is rejected.
- Verify every transition creates an audit event.
- Verify event delivery failure is observable and retryable.

10. Handoff to all persons

Recipient	Must receive
P2	Document schemas, OCR event payload, source/reference rules
P3	Canonical extracted fields, validation issue schema, workflow statuses
P4	GeoAI layer/feature schema, topology status, GeoAI completion event
P5	API contract, workflow tables, auth/RBAC integration points
P6	All API endpoints, response schemas, UI states and permission behavior

11. Definition of done

- All contracts are frozen and versioned.
- Both workflows are executable end-to-end with mocks.
- Illegal state transitions are rejected.
- Retry is idempotent.
- Integration failures are observable.
- Cross-module contract tests pass.
- Release checklist is complete.